

Title: Predicting Antidepressant Toxicity through Graph Neural Network Integration of Chemical Structures and Clinical Reports

Short Title

Predicting Antidepressant toxicity and ADRs using GNNs

Author: Anastasia Levina¹

Affiliation

¹University of Washington, Bothell

Abstract

The use of antidepressants has increased significantly in recent decades, revealing both their benefits and the prevalence of adverse drug reactions (ADRs). While major achievements have been made in personalized pharmacogenomics, current predictive models often rely exclusively on patient genotype data, leaving out the structural properties of the drugs themselves, creating a significant gap. In this paper, I test the hypothesis that a multimodal late-fusion graph neural network can effectively integrate chemical structural data (SMILES string) with pharmacogenomic data to accurately predict a toxicity rating associated with a specific antidepressant. To test this, I developed a data pipeline that extracts molecular properties of drugs using PubChem, filters clinical annotations from the ClinPGx database, and integrates adverse event reports from the openDFA API. This data was then compiled into a graph-based dataset to train a custom graph neural network (GNN). Preliminary evaluation of the model indicated that while the underlying pipeline architecture was successful, the actual outputs converged around a single value, thus not giving a clear representation of the true differences in toxicity for different inputs. This divergence from the expected linear variations suggests constraints in the toxicity rating scaling logic under localized data compression. Thus, this paper establishes a baseline for the multi-modal machine learning integration of chemical and clinical data, demonstrating its general architectural feasibility.

Introduction

Antidepressants are currently one of the most prescribed medications for US adults. As of 2026, 1 in 6 US adults report undergoing antidepressant treatment (Perlis et al., 2026). While antidepressants tend to be extremely effective, there are also many concerns about their side effects, as those often vary drastically from person to person, so it is common for doctors to switch out drugs multiple times before they find the one that actually works for the patient. This

guess and check approach is addressed in a study that examined remission rates for patients taking antidepressants, where at each step if the patient wasn't satisfied with the initial drug, it was switched to a more aggressive one. The theoretical remission rates after the 4 switches of the drugs were an increase from 37% to 67% (Pigott et al., 2023). However, the actual remission rates were significantly lower, with an increase from 30.2% to 41.3% based on medical records, and an increase from 25.5% to 35% based on independent reanalysis (Pigott et al., 2023). This clearly demonstrates that the guess and check method for antidepressant treatment is ineffective.

The primary way antidepressants function is by blocking the reuptake of neurotransmitters, such as norepinephrine, dopamine, and serotonin, which results in them flooding the synapse and enhancing the patient's mood (Lattimore et al., 2005). The main reason why people react differently to antidepressants is because some people's enzymes are less efficient in metabolizing the drugs, which leads to high concentrations of the drug to build up in the body, leading to side effects. While others' enzymes are more efficient in metabolizing the drugs, which results in lower drug concentrations and fewer side effects (Zanger & Schwab, 2013). While this is a very common explanation, it is not the only one, and there are many other complexities that influence the body's response to a drug. Managing these reactions remains a major challenge due to the ambiguity and high variance of reported side effects. To control these risks, computational tools must be developed to aid in the prescription process, protecting patients from incompatible drugs before they are introduced to them.

Recent advancements in pharmacogenomics have introduced AI-based decision support tools, such as Predictix, which uses a patient's genetic data and clinical features to help doctors prescribe the right medication for major depressive disorder (Saudreau et al., 2022). However, a significant gap remains because the majority of these models like Predictix rely primarily on the patient's genotypes, treating the drugs as simple categorical text rather than processing their specific chemical structures, which are crucial to understanding how a drug behaves under specific conditions. Conversely, new research has started using multi-modal graph machine learning to analyze complex biological networks and raw molecular fingerprints to predict generic drug toxicities (Krix et al., 2023). Building upon these frameworks, this project aims to address the gap by connecting specific structural molecular patterns to real-world population safety data.

Antidepressants are categorized into distinct classes, such as Selective Serotonin Reuptake Inhibitors (SSRIs), Serotonin-Norepinephrine Reuptake Inhibitors (SNRIs), Tricyclic Antidepressants (TCAs), and Monoamine Oxidase Inhibitors (MAOIs). Because drugs within these classes share a lot of their chemical structure, they can be represented using the Simplified Molecular Input Line Entry System (SMILES).

In this paper, I test the hypothesis that a multimodal late-fusion graph neural network can effectively integrate chemical structural data (SMILES string) with pharmacogenomic data to accurately predict a toxicity rating associated with a specific antidepressant.

To test this hypothesis, I developed a data pipeline that extracts molecular structures of 20 different antidepressant drugs using PubChem, processes clinical annotations from the ClinPGx database, and combines the real safety reports from the openFDA API. Using PyTorch Geometric, these multi-modal inputs were fused into a unified dataset of thirty clinical interaction profiles to train a GNN capable of mapping non-linear chemical and biological patterns. The architecture was designed to recognize features across the combined inputs to dynamically scale an openFDA-based baseline toxicity score into a final toxicity rating between 0 and 100. Additionally, the output included the top five historically reported side effects associated with the drug. Preliminary evaluation of the model indicated that while the underlying pipeline architecture was successful, the actual outputs converged around a single value, thus not giving a clear representation of the true differences in toxicity for different inputs. This divergence from the expected linear variations suggests constraints in the toxicity rating scaling logic under localized data compression. Thus, this paper establishes a baseline for the multi-modal machine learning integration of molecular and clinical data, demonstrating its architectural feasibility

Methods

The first step of the project is gathering data from the three primary sources: PubChem, ClinPGx, and openFDA. Because these datasets are very large and not specific to antidepressants, I used Python to filter and organize the data for the final model.

To gather the chemical structure of the drugs, I used the PubChemPy library (Kim et al., 2021). I created a list of 20 antidepressant drugs based on classes identified by the U.S. Food and Drug Administration (2019), including SSRIs, SNRIs, TCAs, MAOIs, NMDA. I also included several atypical antidepressants, and some newer drugs, such as NMDA receptor antagonists (e.g. Esketamine) and neuroactive steroid gamma-aminobutyric acid (GABA) receptor positive modulator (e.g. Brexanole). While I tried to include at least three representations of each type of drug, the newer and the atypical drugs only had one to two representations. Using the connectivity_smiles method and a for loop, I retrieved the SMILES string for each drug. I stored this information in a Pandas data frame to create an organized table of drug names and their corresponding chemical structures titled antidepressant_smiles.csv (Supplementary Data 1).

For the pharmacogenomic data, I combined two separate source files from the ClinPGx database (summary_annotations.tsv and summary_ann_alleles.tsv) (Whirl-Carrillo et al., 2021). The primary metadata file (summary_annotations.tsv) was initially filtered to isolate entries containing any of the 30 targeted antidepressants, restricted to evidence levels 1A through 3 within the “Toxicity” and “Metabolism/PK” reaction phenotype categories. These phenotypes were most relevant to my main focus on how antidepressants result in side effects by either being present in high concentrations or being metabolized inefficiently.

109 Additionally, I developed a natural language processing method to process the raw text blocks
110 within the corresponding file (summary_ann_alleles.tsv) to compute a numerical risk_score, which
111 was the primary clinical data. The risk score was based on this criteria:

- 112
- 113 • High risk (2.0): Assigned if the annotation text contained keywords: “increased
114 risk”, “poor metabolizer”, “higher risk”, “increased severity”, “increased chance”
- 115 • Intermediate risk (1.0): Assigned if the annotation text contained keyword:
116 “intermediate” or “reduced”
- 117 • Low risk (0.0): Assigned if none of the above keywords were present

118 Once this scoring was computed, an inner merge was executed across both datasets using the
119 ‘Summary Annotation ID’ as the relational marker. I then exported this data into the
120 annotations_data.csv file (Supplementary Data 2), which contained the columns:

- 121 • Drug(s): The name of the antidepressant
- 122 • Gene: The specific gene governing the metabolic or toxicity pathway (e.g.
123 CYP2D6)
- 124 • Variant/Haplotypes: The broader molecular or chromosomal sequence markers
125 associated with the metabolic group (Note, while this was included in the initial
126 reference file, I wasn’t able to incorporate it into the final model)
- 127 • Level of Evidence: The clinical evidence score from 1A to 3
- 128 • Genotype/Allele: The specific nucleotide or structural allele combination
129 extracted during the two-file merge (Note, this was also included in the initial
130 reference file, and was used for extracting the raw annotations text, but it was not
131 used in the final model training)
- 132 • Risk_Score: The calculated numerical score (0.0-2.0) representing the severity of
133 clinical description
- 134 • Annotation Text: The raw, unprocessed qualitative textual description detailing
135 the observed patient outcomes and pharmacological phenotype

136

137 For the openFDA data processing, I used the openFDA API to retrieve clinical safety reports
138 (U.S. Food and Drug Administration, 2025). I focused on two specific fields: the seriousness
139 level of the report and the specific standardized term describing the reaction associated with the
140 drug. These terms are standardized using the Medical Dictionary for Regulatory Activities
141 (MedDRA) under the variable “reactionmeddrapt” in the files. Before gathering the data, I
142 developed a ‘skip’ list to filter out terms that were not specific enough for my project. I
143 programmed the code to skip over terms like “pain”, “off label use”, “suicidal attempt”,
144 “depression”, etc. While these are valid side effects, I wanted to focus specifically on physical
145 and physiological effects (e.g. nausea, tremors, etc.) that directly related to how a drug’s
146 chemical structure interacts with a patient’s genetics, rather than a person’s direct actions. Using

Python Requests library, I queried the API for the top one hundred most frequent reactions for each drug in my list with a seriousness level of 1 (serious:1). The script looped through the drugs, checked for available data, and stored the drug name, the reaction term, and the reaction count. It stored a total of fifty (accounting for the noise data). Finally, I converted this data into a Pandas data frame, similar to the SMILES and ClinPGx data, titled openfda_data.csv (Supplementary Data 3).

The next step was converting the SMILES string into a graph that could be fed into the model. In order to do this, I made a function called `convert_smiles_to_graph`, using the RDKit library. This function processed a SMILES string and converted its chemical structure into a PyTorch Geometric data object, where the atoms were the nodes, and the bonds were the edges. For the nodes, the computer cannot interpret letters, so the code got the atomic number for each element in the molecule and stored them in a node feature tensor x . For the edges, each atom in the molecule was given an index ID (0, 1, 2, etc.). The final connection was stored into an `edge_index` matrix, where the top row held the starting index of a bond, and the bottom row held the ending target index. Since chemical bonds do not have a direction, each bond was added twice, switching the order of the source and target nodes. Additionally, I created an `edge_attr` tensor that kept track of the specific bond type by assigning it a double value based on the bond type, where a single bond is 1.0, a double bond is 2.0, a triple bond is 3.0, and an aromatic bond is 1.5.

Next, to bridge the chemical and clinical data before training the model, I wrote a fusion script to make the final multi-modal dataset, as shown in the data integration path in workflow chart (Figure 1). First, the code processed the safety reports by summing up all serious adverse events per drug from `openfda_data.csv`. To standardize these numbers, I applied a min-max normalization formula (Thafar et al., 2021). This scales each drug's aggregate report volume (C_d) against the absolute minimum (C_{min}) and maximum (C_{max}) counts observed across the entire dataset, constraining the baseline toxicity rating (B_d) on a 0.0 to 100.0 scale:

$$B_d = \left(\frac{C_d - C_{min}}{C_{max} - C_{min}} \right) \cdot 100$$

Next, the loop iterated through the rows of `annotations_data.csv` to combine these baselines with the risk score data. If a row's risk score was intermediate (1.0) or high (2.0), the script scales the baseline score by an additional 25% or 50% respectively. There is also a ceiling logic to ensure that the final y does not exceed 100.0:

182
$$y = \min(B_d \cdot (1.0 + (R \cdot 0.25)), 100.0)$$

183 To prevent cross-contamination across profiles, a deep copy of each drug's graph is created
184 before appending these targets. Additionally, the script creates a dense genetic attribute tensor
185 (genetic_attr) by building a fixed-size index of all unique genes to generate a one-hot encoded
186 vector, which is then concatenated with the risk score. This combined dataset is the complete
187 multi-modal training data for the model.

188 The, to handle the structural data, I built a custom network class called MultimodalGNN that
189 processes the chemical graphs and patient genetics together (Figure 1). The branch consists of
190 three graph convolutional layers (GCNConv) designed to pass feature information across
191 neighboring atoms. The first layer compresses the raw atomic number inputs into a 16-
192 dimensional hidden layer; the second layer expands this representation to 32 dimensions, and the
193 third later outputs a final 64-dimensional feature vector for every atom in the molecule. To
194 expand upon single-modality frameworks that isolate genotype data (Saudreau et al., 2022). To
195 combine this chemical path with the clinical data, I calculated a fixed size for the patient vector
196 (genetic_input_dim) equal to the one-hot encoded gene array plus the risk score. I then defined
197 three connected linear layers (nn.Linear) to process the merged data, shifting the combined
198 feature vector down from its initial size to 32 hidden nodes, then to 16, and finally down to a
199 single output value (Figure 1).

200 When executed, the model's forward function unpacks the atom feature tensors, bond edge
201 indices, and tracking IDs, along with the clinical tensor (genetic_arr). The atom data is pushed
202 one by one through the three graph convolutions using Rectified Linear Unit (ReLU) activation
203 functions to map the non-linear atomic geometry. Once the atom features are computed, the
204 model uses a global_mean_pool layer to average the node features across each molecule,
205 compressing the entire 2D structural layout into a single 64-dimensional drug fingerprint.
206 Building off the geometric deep learning techniques that map raw chemical network to adverse
207 outcomes (Krix et al., 2023). For the late-fusion integration, the script uses torch.cat to bind this
208 chemical fingerprint directly with the patient's genetic vector. This combined row passes through
209 the linear layers using ReLU activations to compute the final drug-genotype interactions. Finally,
210 the raw network output is passed through a Sigmoid function and multiplied by 100.0, scaling
211 the final predicted toxicity rating to the 0.0-100.0 scale (Figure 1).

212 The final part was enabling the program to take input and use the model network to generate live
213 outputs. To do this, I made the user_input function that gets two distinct inputs from the user: the
214 drug name and text describing the genetic characteristics of the patient. Theoretically, these
215 inputs are supposed to be the active gene and a description of the patient's metabolic process.
216 The function parses this text using the keyword script, using similar, but slightly expanded
217 categories as the initial data processing code. The code finds the gene name if it is present and
218 calculates the risk score based on the input. Next, the function looks up the drug entered by the
219 user in the database to retrieve its molecular graph. It then updates the graph by attaching the risk

vector directly to the genetic_attr tensor. This combined data is passed into the MultimodalGNN network to run a forward pass. The model processes the inputs and outputs a scaled toxicity rating between 0.0 and 100.0, while simultaneously looking up and printing out the top 5 most frequent real-world side effects for the specific antidepressant.

Results

To test my hypothesis, I performed a 2-sample t-test, comparing the means of the baseline toxicity scores group and the toxicity scores that the model predicted. The baseline toxicity score in this case is the approximation of the true toxicity score for each input. I used a sample of 30 distinct inputs (which included the drug name and different descriptions of the genetic characteristics, and for each input, I recorded the baseline score by adding code that extracts and prints it in the user_input function. My null hypothesis was that there is no difference between the means of the baseline and the predicted toxicity scores, while my alternative hypothesis was that there is a difference between the means of the baseline and the predicted toxicity scores:

$$H_0: \mu_{\text{model}} = \mu_{\text{baseline}}$$

$$H_a: \mu_{\text{model}} \neq \mu_{\text{baseline}}$$

After running the test using a graphing calculator, I got a test statistic of $t = 0.6176$ and a p-value of 0.5416. Since the p-value is greater than any reasonable significance level ($0.5416 > 0.01$ and $0.5416 > 0.05$), we fail to reject the null hypothesis and can conclude that there is insufficient evidence to suggest that there is a difference between the means of the baseline toxicity scores and the toxicity scores predicted by the model. However, since the overall means matching is due to the model predicting a flat average value, these results suggest that I am unable to conclude that this specific implementation of a multi-modal late-fusion GNN can accurately predict toxicity scores associated with specific antidepressant drugs.

Additionally, in order to visualize my results, I created a plot that maps the compares baseline toxicity on the x-axis scores against the model's predicted toxicity scores on the y-axis. The plot contains a solid black line with a slope of 1, which represents the identity of line where the prediction would land if the results were perfect ($y=x$) (Figure 2). The baseline and predicted toxicity scores are combined into a single coordinate and represented as blue dots. The plot was made using Python, specifically using the Matplotlib and Seaborn libraries. When plotting these 30 outputs, the resulting blue data points created a horizontal cluster centered around the 43.0 mark on the y-axis (Figure 2). This suggests that the model's outputs converge toward the mean baseline value. The plot explains the results of the 2-sample t-test, demonstrating that while the overall means are the same, the model is experiencing mean convergence and underfitting instead of matching the individual baseline toxicity scores across specific inputs.

Discussion

Based on the results of my 2-sample t-test, since the p-value (0.5416) was significantly higher than the significance level of 0.05, I failed to reject the null hypothesis that there is no difference between the mean toxicity scores of the baseline group and the predicted toxicity scores. Statistically, this means that the overall averages of the 2 groups are the same, which would mean that the predicted values are equal to the true values. However, because the plot showed that the model predicted a flat value for nearly every input instead of changing it based on the input data (Figure 2), it means that the model was not able to accurately predict the true toxicity score for the inputs. This means that both the baseline and the predicted toxicity scores were calculated inaccurately. At this point, I also reject my main hypothesis that this multi-model late-fusion model implementation can accurately predict the toxicity score for a specific antidepressant and genetic input.

The reason the model's output converged around 43.0 was likely due to the neural network minimizing its loss function during training. Because the dataset was small and the raw openFDA target scores were not very accurate, the model resulted in underfitting. The network 'figured out' that the safest way to reduce its overall training error without understanding the deep difference between the drug structures was to split the difference and repeatedly output the global average score, which was around 43.0. This resulted in the model predicting almost the same scores for every input.

While the individual predicted scores diverged extremely from the true changing toxicity scores, I would not completely suggest that the model itself was completely inaccurate since throughout the runs with different inputs, I could see that the model was able to recognize patterns based on the supplementary output it provided, such as correctly identifying the risk score, the gene, and creating the molecular graph, and overall, the model seemed successful in combining the different data types and datasets it was given.

In my opinion, the major problem with the results was the way of calculating the baseline scores since they relied solely on the number of adverse reports from openFDA (Figure 1). While this data is accurate and credible, it is also quite skewed. Since some of the drugs I trained the model on were newer (e.g. Esketamine or Brexanolone), there were very few reports on them in openFDA, which would assume the toxicity score to be 0.0. On the other hand, older drugs that have been around longer might have had much more reports, which would lead to an assumed toxicity score that might be significantly higher than the real toxicity score. Thus, I believe that this is one of the major flaws of my methodology, and I should have used a different way of calculating this baseline score, potentially using the total number of sales of the drug in the US and applying the adverse drug event report data on it to find the true rate.

Overall, my results diverged from similar works, where the multi-modal frameworks often proved to be highly accurate. For example, in the study discussed earlier where researchers

implemented a deep learning pipeline that mapped molecular graph structures to a database of side-effect frequencies rather than simply calculating these frequencies based on reports like I did in my project. Because their dataset used uniform frequency rates, it allowed their network to find clear boundaries instead of flattening out at a single value (Krix et al., 2023).

In the future, I believe that the predicted score calculation needs to be changed significantly by changing the output type completely. I think that a more feasible way of representing the output is using classification. This is similar to the approach in the paper that was also discussed previously in a study focused on patients with MDD and the use of the model Predictix to aid in the prescription process. In that study, the results were grouped into low, medium, and high relative side-effect frequency classes (Saudreau et al., 2022). I think that this is a significantly more effective way of representing results since it would be more readable for the users, where they can compare drugs directly without having to figure out what a single toxicity score rating means.

Data Availability

The raw .tsv files from the ClinPGx database are available at <https://www.clinpgx.org/downloads> (Summary and Variant Annotations Data)

The openFDA API description is available at <https://open.fda.gov/apis/drug/event/>

The PubChem Python library is available at <https://docs.pubchempy.org/en/latest/api.html>

Supplementary Code

All the code for this project is available at <https://github.com/AnastasChromostas/CSS383-Project>

Supplementary Data Tables

All the supplementary data tables are available in [Supplementary_Data.xlsx](#)

Figures

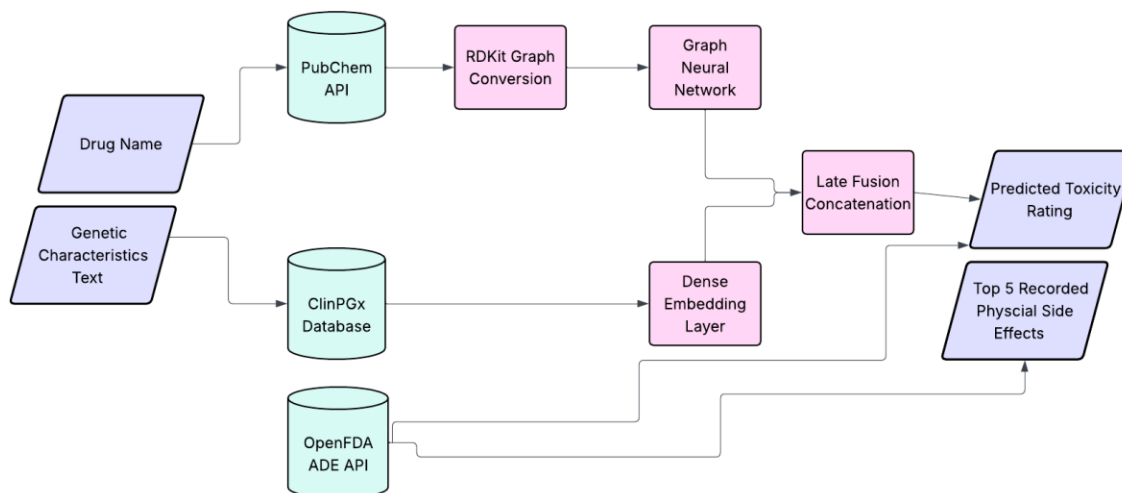


Figure 1. The flowchart of the model logic. The figure represents the surface-level logic of how the model processes inputs and generates outputs. The lavender rhombi on the left represent the inputs, which include “Drug Name” and “Genetic Characteristics Text”. The light mint cylinders represent the databases/data sources, which include “PubChem API”, “ClinPGx Database”, and “OpenFDA ADE API”. The light pink rectangles represent the processes through which the data was processed. The lavender rhombi on the right represent the outputs, which include “Predicted Toxicity Rating” and “Top 5 Recorded Physical Side Effects”. The lines connecting the symbols represent the data flow between the various components.

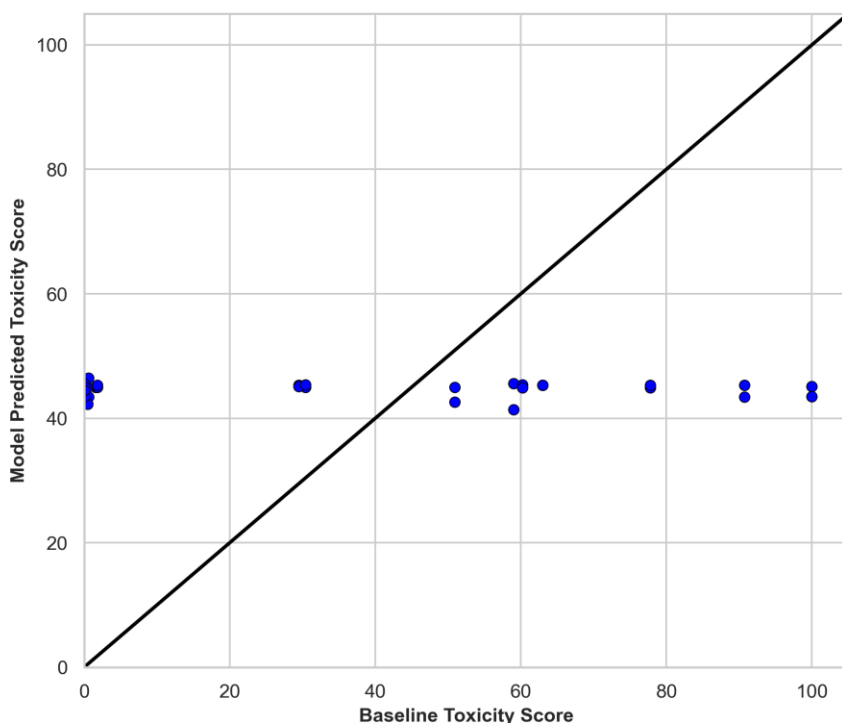


Figure 2. Model prediction alignment plot. The scatterplot maps the baseline toxicity scores calculated using the openFDA reports on the x-axis against the toxicity scores predicted by the model on the y-axis for the 30 inputs used during testing. The diagonal black line represents the identity line ($y=x$), which shows the locations where the coordinates would lie if the predictions were perfect. The blue points represent the model predicted outputs, where each coordinate (x, y) = (baseline toxicity score, predicted toxicity score).

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